

The Σ limit

large the Stieltjes form is exact; the occupation mass
stays off the pole; the limits stay open

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Abstract

Round 421 left a preferred floor $R_\infty \approx 0.03$ and asked whether $\Sigma_k = 2s_k^T(C_k + I)(C_k - I)^{-1}s_k$ stays bounded with $\limsup \Sigma_k < |c_{J,\infty}| = 2 - \text{den}_\infty$. The occupation width n_{eff} grows, so the danger is mass walking into the g -pole at $\lambda = 1$. The Stieltjes identity is exact. An elementary closed occupancy density is refuted. Near-1 mass stays $O(0.01)$. The two limits are not constructed. No RH claim.

Status. Lemma 1 is **reduced**. The Stieltjes / occupation-trace identity is a finite theorem (Proposition 1). A one-pole or two-moment closed density is refuted (Proposition 2). Ausgang STIELTJES_EXACT / DENSITY_REFUTED / NEAR1_STABLE / LIMITS_OPEN. No L^* claim. No $R^\dagger$ claim. No RH claim.

1 The lemma

Lemma 1 (Σ limit). *On the selected sequence $a_k = 2^k$, $\Sigma_k = 2s^T(C + I)(C - I)^{-1}s$ stays bounded with $\limsup_k \Sigma_k < |c_{J,\infty}| = 2 - \text{den}_\infty$, source-pure.*

Verdict: REDUZIERT. $\Sigma = 2\|s\|^2 + 4m_{\text{occ}}(1)$ is exact. Near-1 mass is stable $O(0.01)$: the n_{eff} growth is bulk broadening, not pole-creep. EXT large- N values of Σ stay below 0.43, not above the selected $k = 9$ value 0.480. But den_∞ and Σ_∞ are not constructed (den still wanders in $[1.48, 1.65]$). The inequality remains a census.

2 The Stieltjes identity

Write $g(\lambda) = (\lambda + 1)/(\lambda - 1)$ and $\text{occ}_i = |\langle \psi_i, s \rangle|^2$ for the eigenpairs of C . Then $g(\lambda) = 1 + 2/(\lambda - 1)$, so

$$\Sigma = 2 \sum_i \text{occ}_i g(\lambda_i) = 2\|s\|^2 + 4s^T(C - I)^{-1}s = 2\|s\|^2 + 4m_{\text{occ}}(1). \quad (1)$$

Over $\mathbb{Q}$, $C = \text{diag}(2, 4)$, $s = (1, 1)$ gives $\Sigma = \frac{28}{3}$. On the construction the residual against $s^T A_0^{-1}s$ is 10^{-13} – 10^{-9} .

Proposition 1 (Stieltjes form). (1) holds as a finite identity (r407: $A_0^{-1} = 2(C + I)(C - I)^{-1}$). Dropping the pole ($g \equiv 1$) yields $2\|s\|^2$, which is not Σ (w9: 0.111 versus 0.313).

This is the source-pure form: $m_{\text{occ}}(1)$ is the Stieltjes transform of the occupancy measure at the pole. It is not yet an elementary source sum.

3 No closed occupancy density

Proposition 2 (closed density). *A one-pole ansatz $m(1) \approx \|s\|^2/(\lambda_w - 1)$ misses $m(1)$ by relative error 0.73 on every selected window. A two-moment Cauchy ansatz misses by 0.95. No elementary $\text{occ}(\lambda)$ (in particular, not the derivative of an explicit scalar Stieltjes transform built from λ_w or the first two moments) is obtained.*

The border direction s is a CD-border column. Its overlap with the C -modes is a genuine spectral measure, but the r407 dictionaries do not reduce it to a closed density in this round.

4 Mass near $\lambda = 1$

Selected occupancy mass on $(1, 1 + \delta]$ ($k = 8$ not recomputed; builder pin):

k	Σ	$\mu(1, 1.1]$	$\mu(1, 1.3]$	n_{eff}
5	0.244	0.0003	0.050	18
6	0.408	0.0069	0.073	46
7	0.430	0.0089	0.071	98
9	0.480	0.0108	0.061	146

The $\delta = 0.1$ slot stays $O(0.01)$. The $\delta = 0.3$ slot is flat ~ 0.06 . Quantiles settle ($q_{50} \rightarrow 2.33$, $q_{75} \rightarrow 4.04$). KS distances on the occupancy law fall: 0.140 ($k = 5 \rightarrow 6$), 0.096 ($6 \rightarrow 7$), 0.044 ($7 \rightarrow 9$). That is a weak-convergence *census*, not a theorem. Razor share on living vacuous windows is 10^{-8} – 10^{-10} .

The reading “ n_{eff} growth means mass walks into the pole” is refuted as a mechanism: the extra modes sit in the bulk.

5 The three numbers versus the floor

Pointwise $R = 2 - \text{den} - \Sigma = -\varphi_{bb}$ is the r418/r420 identity. On selected $k = 5, 6, 7, 9$ plus the $k = 8$ pin, $\text{den} \in [1.48, 1.65]$ and $\Sigma \in [0.24, 0.48]$. EXT vacuous $\Sigma \in [0.405, 0.430]$ at $N = 2472, 3022$ — not above $k = 9$. The floor $R_\infty \approx 0.030$ is the residual of two $O(0.5)$ wanderers. It is consistent with every finite row and is *not* a closed triple $(\text{den}_\infty, \Sigma_\infty, R_\infty)$.

If both limits existed with $\Sigma_\infty < 2 - \text{den}_\infty$, the cofinal edge would be that inequality plus the finite census $R > 0$ on the named living windows. Existence of the limits remains the missing piece.

6 Kills

Overflow six: $\mu(1, 1.1] \in [0.010, 0.017]$, systematically above living vacuous ~ 0.007 ; $\Sigma > |c_J|$ at finite N . Dead vacuous χ : razor occupation $1.9 \cdot 10^{-4}$ and $5.7 \cdot 10^{-4}$ (four to five orders above the living vacuous razor). $g \equiv 1$ breaks the identity. Scramble $\text{nneg}(A_0) = 21$. Drop-border: no Σ scalar.

7 What is a theorem, what is a census

Proposition 1 is a theorem. Proposition 2 is a named refutation. Near-1 stability and the falling KS distances are a census. Lemma 1 is not a theorem: boundedness of Σ is observed, the pole is not eating the mass, but den_∞ and Σ_∞ are not built, so $\limsup \Sigma < 2 - \text{den}_\infty$ stays conditional. No RH claim.